MIT Sloan & Conagra 2026 Capstone Project

Manufacturing Changeover Scheduling Optimization

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Conagra Points of Contact: Adrian McClure, Paige Boncosky

MIT Faculty Advisor: Bruce Lawler

Contents

1 Background and Problem Statement

Conagra Brands operates a large, multi-site manufacturing network where production lines frequently switch between SKUs.
Each changeover (cleaning, setup, material swaps, inspections) creates downtime and can drive decreased efficiency, schedule instability, and trade-offs across service and margins. Sequencing decisions are often driven by planner experience and fragmented information across systems, making it difficult to consistently identify the optimal feasible sequence under real-world constraints such as allergen segregation, sanitation requirements, equipment compatibility, color changes, service priorities, and inventory risk. Current solutions to generate schedules and adjustments are completed manually and not standardized across plants.

2 Objective

Build a changeover tool to output a detailed schedule that optimizes service, fill rate, changeover cost, and manufacturing efficiency, while incorporating an agentic solution for interpretability and responding to unexpected events in real time.

3 Scope, Data Sources, and Tools

3.1 Scope:

The initial phase will focus on developing and piloting a plant-level sequencing optimization solution designed to improve manufacturing efficiency, reduce changeover costs, and enhance fill rate. This initial formulation phase will validate technical feasibility and stakeholder usability. The pilot will emphasize practical integration into planner workflows, combining algorithmic recommendations with transparency and explainability to ensure adoption and trust.

- Pilot at 1 pilot site (Fayetteville) **(no longer Counsil Bluffs site)**
- Sequencing within a defined planning horizon (shift/day/week)
- Agentic built planner-facing recommendations with explainability
- Offline back testing and pilot evaluation

3.2 Primary Data Sources:

The solution will integrate operational, manufacturing, and compliance data to create a sequencing engine that balances efficiency, quality, and feasibility constraints. Data will be structured to reflect real production constraints, regulatory requirements, changeover costs, and ingredient availability, ensuring recommendations are both feasible and aligned to business KPIs. Historical changeover data will support model training and feasibility assessment, while real-time planning data will test schedule outputs and manual overrides. Data sources are on BlueYonder, SAP platforms, **Databricks, and Excel**.

- SKU master attributes (allergens, product family, packaging type, color class)
- Historical production and changeover records

- Quality and sanitation compliance rules (e.g., allergen protocols)
- Equipment setup and centerline data
- Cost and margin signals (SAP production cost data)
- Order demand, inventory, and service data
- **Labor requirements and availability**
- **Finished good material dependencies**

3.3 Tools and Platforms:

- Databricks for data engineering (Bronze, Silver, and Gold) and modeling (MLflow)
- SAP PPDS and possibly Palantir for planning/action integration
- Power BI for reporting and visualization dashboards
- Python-based optimization stack (heuristics, MILP)
- Agentic builds in Databricks Agent builder
- **Excel Sheet for changeover dashboard**

4 Interval Communication (Daily Points of Contact)

Daily

Adrian McClure; Paige Boncosky; Davis Fulk (through May; rotates off the team in June).

Weekly / As Needed

Alfred Paige; Daniel Hitchcock (Fayetteville Plant); Alicea Norton (Planning Center of Excellence Lead), **Robert Dorch (Fayetteville Plant Data Expert); Ella Knee (Fayetteville Plant Data Expert); Vanessa Leedom (Fayetteville Schedule Planner)**.

5 Deliverables, Success Criteria, and KPIs

5.1 Deliverables:

The pilot will produce a set of production-ready scheduling tools that enable data-driven sequencing decisions and establish a foundation for scalable deployment with the expansion of data uniformity across plants. Deliverables will span data engineering and suggestions for company-wide metrics, predictive modeling, optimization development, workflow integration, and performance evaluation. Together, these outputs will demonstrate both technical viability and operational readiness for broader rollout.

- Unified changeover dataset with documented schema
- Changeover cost prediction model (rules + ML)

- Constrained sequencing optimizer
- Agentic orchestration layer integrated into planner workflow
- Evaluation report and pilot rollout plan

5.2 Success Criteria:

Success will be defined not only by algorithmic performance, but by operational usability and measurable business impact. The solution must generate recommendations that are feasible within real-world plant constraints, transparent to planners, and demonstrably superior to historical sequencing practices. Additionally, the system must operate within existing planning cadences and drive meaningful planner engagement and trust. Planner adoption must be feasible and adjustable with real-time constraint changes.

- Operationally feasible and explainable recommendations
- Measurable improvement over historical sequencing **(efficiency improvement, % of schedule adjustment improvement)**
- Runtime compatible with planning cadence
- Planner adoption and low override rates

5.3 Key Performance Indicators (KPIs):

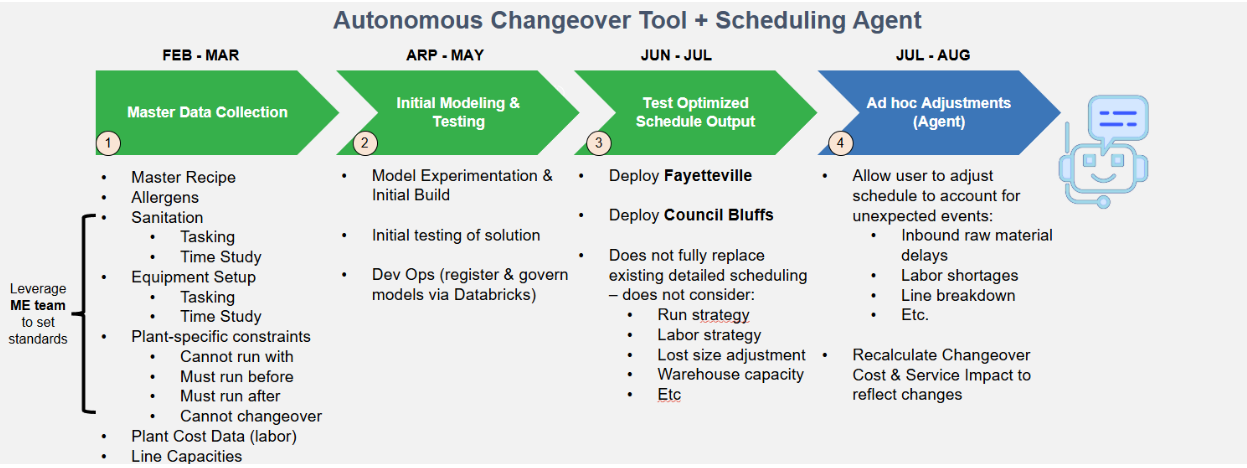
Performance will be evaluated using quantitative metrics that capture efficiency, service, stability, and system reliability. These KPIs are designed to assess both direct operational improvements and broader planning performance outcomes. Measurement during the pilot will establish a benchmark for scaling and long-term value.

- Total changeover time reduction (total equipment downtime + ~~**total labor hours**~~)
- Fill rate
- Schedule stability and constraint compliance
- Model accuracy override percentage vs. Detailed schedule
- System runtime and robustness
- Improved efficiency (actual uptime / planned uptime)

6 Estimated Project Timeline

During February, March, **and April**, data collection and exploratory analysis will take place. After ensuring the data are in the correct format and fully understood, initial modeling and development will begin in April. In May, the goal is to test the initial solution so that it can be refined and made ready for deployment at the Fayetteville site during June and July. The deployed system will serve as an interface for end users rather than an auto-pilot. By late July and early August, the system will enable users to adjust schedules in response to unexpected events, and the team will deliver a final evaluation of overall business impact and KPIs for the delivered system.

This timeline is summarized in the graphic below **(Counsil Bluffs is not longer a part of the pilot roll out)**:



7 Potential Limitations, Constraints, and Risks

7.1 Data Risks:

The quality and completeness of historical production data directly influence model reliability and optimization accuracy. Inconsistent timestamping, limited observations for rare transitions, and unobserved operational factors may introduce noise or bias into cost estimation and sequencing recommendations.

- Noisy or incomplete changeover timestamps
- Sparse data for rare transitions (cold start)
- Confounding factors such as crew or maintenance effects

7.2 Operational Risks:

Operational feasibility is critical to success. Optimal solutions may fail if they do not reflect real-world plant dynamics or if planners lack trust in the recommendations. Change management and stakeholder engagement will therefore be central to adoption.

- Incorrect encoding of hard constraints (e.g., allergen rules)
- Recommendations that are optimal but impractical
- Resistance to adoption without transparency

7.3 Technical Risks:

- Integration complexity with SAP and planning systems

- Latency constraints for real-time use
- Data leakage during model development

7.4 Mitigation Strategies:

- Conservative rule-based fallbacks
- Early validation with QA and operations
- Clear explainability and override logging
- Agentic explainability layer

8 Project Dependencies

Successful execution of the pilot is contingent upon foundational dependencies across data access, governance alignment, and stakeholder engagement. Early clarity on planning parameters, constraint definitions, and output schedule format will be critical to avoid rework during model development. In addition, consistent plant and scheduler involvement and platform access are required to ensure technical progress and operational validation remain on track.

- Access to historical production and changeover data
- Clear definition of planning horizon and decision timing
- Formal agreement on hard vs. soft constraints
- Baseline sequencing benchmark for comparison
- Subject matter expert availability for validation
- Secure access to required data platforms

9 Project Stakeholders

9.1 Business Stakeholders:

- Plant leadership and production management
- Supply chain planning and scheduling teams
- Manufacturing excellence team

9.2 Technical Stakeholders

- Supply Chain Analytics and Data Science Team (SCANDS)

9.3 Risk and Compliance:

- Quality assurance and food safety
- Regulatory and audit stakeholders

9.4 End Users:

- Line schedulers, supervisors, and operators, Plant Leadership, Detailed Schedulers, and supply planners

10 Appendix: Kickoff Meeting Notes

10.1 Meeting 1: Initial Conagra Kickoff

Date: February 9, 2026
Attendees: Student Team, Adrian McClure (Director of Advanced Analytics, Sponsor)

Agenda

- Align on project logistics and administrative setup
- Discuss hardware provisioning and systems access
- Establish communication cadence

Discussion

- Discussed hardware logistics and the process for assigning company laptops to the student team.
- Initiated the process for internal systems access provisioning.
- Discussed the format and frequency of recurring check-ins.

Decisions

- Weekly sync call scheduled for Fridays.
- Travel plan established: Team will spend two months in Chicago, with 3 days on-site per week.

Open Questions

- Exact start & end date for on-site work in Chicago to be confirmed.

Action Items

Action Item	Owner	Due Date
Deliver company laptops to student team	Adrian / Conagra IT	Feb 20
Process internal systems access approvals	Adrian / Conagra IT	Feb 20

10.2 Meeting 2: Initial MIT Advisor Meeting

Date: February 11, 2026
Attendees: Student Team, Bruce Lawler (Faculty Advisor)

Agenda

- Introductions between team and faculty advisor
- Project overview and context-setting
- Discuss data strategy considerations

Discussion

- Backgrounds shared between the team and advisor.
- Discussed the project scope; Bruce shared insights from a similar project he worked on early in his career.
- Bruce stressed that data suitability and quality are much more critical than having a massive volume of data.

Decisions

- Prioritize early access to data to assess quality before building any models.

Open Questions

- What is the current state of Conagra's data quality?
- What specific data recommendations should be made once data is reviewed?

10.3 Meeting 3: Conagra Touchpoint with RL Rosqueta

Date: February 18, 2026
Attendees: Student Team, Adrian McClure (Director of Advanced Analytics), Paige Boncosky (Digital Product Manager), RL Rosqueta (Supply Chain & AI Thought Leader)

Agenda

- Introductions with RL Rosqueta
- Discuss project business impact and practical applications

Discussion

- First meeting with RL Rosqueta; introductions completed.
- Focused heavily on the project's expected business impact and practical applications.
- RL emphasized that fully understanding the project will require specific supply chain expertise and scheduler context from SMEs.

Decisions

- SME engagement is required to ground the project in operational reality.

Action Items

Action Item	Owner	Due Date
Identify specific SMEs for project engagement	Adrian / Paige	Feb 28
Schedule initial SME sessions	Paige Boncosky	Mar 7
Prepare SME questions/ Problem Formulation	Student Team	Mar 7

10.4 Meeting 4: Conagra Weekly Touchpoint

Date: February 20, 2026
Attendees: Student Team, Adrian McClure (Director of Advanced Analytics), Paige Boncosky (Digital Product Manager)

Agenda

- Complete laptop setup and hardware logistics
- Share initial project charter draft for feedback

Discussion

- Company laptops arrived; team worked through the initial logistics of logging in and setting up hardware.
- Shared a brief draft of the project charter with Adrian and Paige to ensure alignment on goals.

Decisions

- Received initial, high-level feedback on the drafted charter. Sponsor team to provide detailed feedback offline.

Open Questions

- What specific areas of the charter need refinement per the sponsor team's detailed review?

Action Items

Action Item	Owner	Due Date
Review charter offline and provide detailed feedback	Adrian / Paige (Sponsor)	Feb 27
Complete laptop setup and verify access	Student Team	Feb 27

10.5 Meeting 5: Conagra Weekly Touchpoint (Expanded SCANDS Team)

Date: February 27, 2026
Attendees: Student Team, Adrian McClure (Director of Advanced Analytics), Paige Boncosky (Dig-

ital Product Manager), Alfred Paige (Senior Data Scientist), Davis Fulk (Supply Chain Development Associate)

Agenda

- Review sponsor feedback on project charter
- Clarify project scope, objectives, and constraints
- Finalize travel plan

Discussion

- Reviewed the major feedback provided by the sponsor team on the project charter.
- Clarified the project's main objectives and constraints to ensure both the team and sponsors are fully aligned.

Decisions

- Project scope, objectives, and constraints clarified and agreed upon.
- Travel plan finalized, locking in the exact dates the team will be physically on-site in Chicago.

Alignment Assessment

Strong alignment achieved. Both teams agreed on scope, objectives, constraints, and travel logistics. Charter effectively finalized after incorporating sponsor feedback.

Action Items

Action Item	Owner	Due Date
Share finalized charter for sign-off	Student Team	Mar 3
Return the charter signed-off	Adrian / Paige (Sponsor)	Mar 7

10.6 Meeting 6: HR/Technology Onboarding

Date: March 2, 2026
Attendees: Student Team, Adrian McClure (Director of Advanced Analytics), Paige Boncosky (Digital Product Manager)

Agenda

- Technical onboarding to Conagra platforms
- Discuss data pipeline architecture and project integration
- Knowledge transfer on Conagra/Databricks terminology

Discussion

- Began logging into internal platforms: Databricks and Azure DevOps.
- Discussed Conagra's existing data pipeline architecture and exactly how this project will integrate into it.
- Learned essential Conagra and Databricks jargon to streamline future communication.
- Held initial discussions about the specific development tools required for the build phase.

Open Questions

- Are there any access permissions still pending for Databricks or Azure DevOps?

Action Items

Action Item	Owner	Due Date
Familiarize with development tools for build phase (Databricks and Azure DevOps)	Student Team	Mar 7
Confirm all access permissions are in place	Adrian McClure	Mar 7

10.7 Project Scope Amendment: Pilot Site Refinement

Date: April 3, 2026
Communication from: Paige Boncosky

Update on Scope

On April 3, 2026, the Conagra project hosts informed the MIT Sloan team of a strategic adjustment to the pilot scope. Due to an overlapping data initiative currently underway at the Council Bluffs facility, it was determined that the site would be unable to provide the necessary operational support and data readiness required for this project within the established 2026 timeline.

Impact on Project Execution

- **Consolidated Focus:** The project will transition from a dual-site pilot to a single-site deep dive focusing exclusively on the **Fayetteville Plant**.
- **Resource Allocation:** Engineering and modeling efforts will be concentrated on Fayetteville's specific constraints to ensure a high-fidelity, production-ready solution.

11 Work & Travel Plan

This Work & Travel Plan outlines the operational structure and schedule for the project team during the Spring and Summer 2026 terms.

11.1 Term Schedules & Travel Details

Spring Term: February 22 – May 22
Format: Remote.
End Date: May 22, 2026 (aligned with the MIT Spring semester).
Travel: Optional site visits to Company offices or manufacturing plants may be scheduled to deepen project context and strengthen stakeholder relationships.
Interim Availability (March 16 – March 27): During the Sloan Independent Activities Period (IAP) and Spring Break, the student team will have limited availability. While the team will remain responsive to messages and available for brief synchronization calls, project-intensive work and deliverables will be paused during this two-week window.

Summer Term: June 8 – August 14 (10 Weeks)

Dates / Location / Format

Dates	Location	Format
June 8 – August 7	Chicago, IL	Hybrid: Minimum 3 days/week on-site at Merchandise Mart Plaza Office.
August 10 – August 13	Boston, MA	Remote: Final preparation for the showcase.
August 14	Boston, MA	On-site: Capstone Showcase.

Transition Period: May 23 – June 7
Format: Daily touchpoints.
Objective: The student team will remain in daily contact with the Company to maintain momentum and ensure full preparation for the start of full-time summer work.

11.2 Summer Travel Logistics

The student pair will relocate to Chicago for a 2-month residency to facilitate high-touch collaboration with the host team.
Travel Destination: Chicago, IL.
Chicago Arrival: June 8, 2026.
Chicago Departure: August 7, 2026.
Office Location: Merchandise Mart Plaza, Chicago.
Final Week: The team will return to Boston on August 7th. The final week of work (August 10–14) will be virtual, culminating in the Capstone Showcase in Boston on the final Friday.